

Official Pipeline: Gaze Selects + EMG Confirms

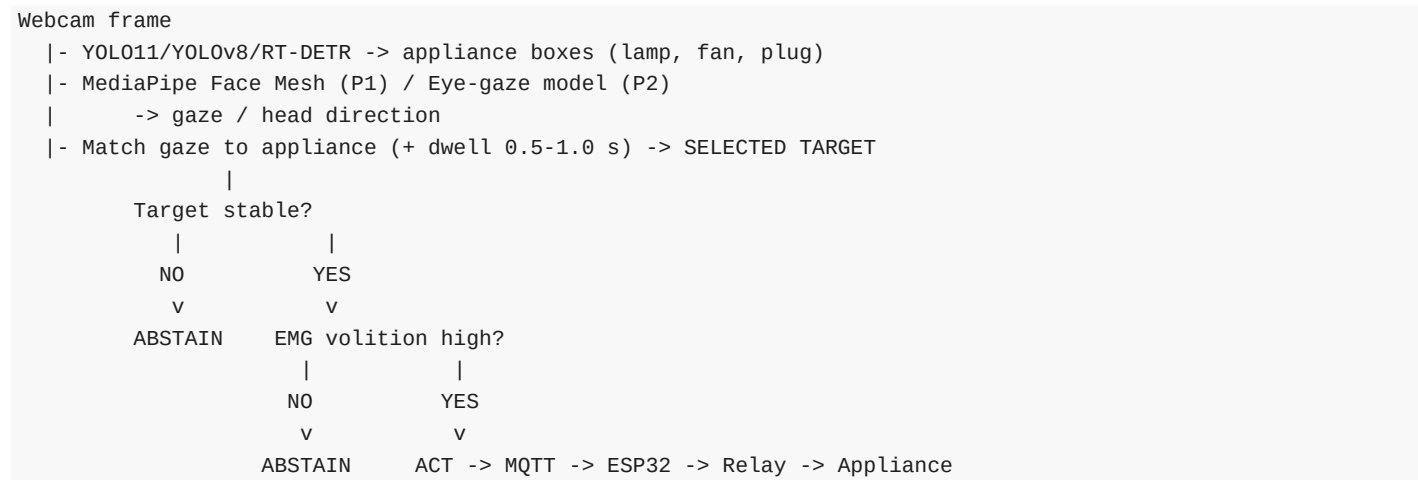
Updated NeuroShift real-time design for Phase 1 and Phase 2 | July 2026

Looking selects. Muscle confirms. Unsure -> Abstain. Gaze alone must never turn a device on or off.

1. Why this design

- Camera finds appliances AND which one the user is looking at.
- EMG confirms whether they truly want to toggle it.
- Prevents Midas-touch: looking without intent does nothing.
- Better for motor-impaired users: eyes often work when hands do not.
- Stronger novelty vs base paper (hand/object grip validation).

2. Real-time pipeline



3. Two-factor safety table

Signal	Role	Alone enough to actuate?
Gaze / head direction	SELECT appliance	NO
EMG effort	CONFIRM intent	NO (needs selected target)
Gaze + EMG together	Complete intention	YES -> Act
Either missing / unclear	Unsafe	ABSTAIN

4. Phase 1 vs Phase 2 gaze

Phase	Gaze method	Why
Phase 1 (Sem 7)	Head/face orientation (MediaPipe Face Mesh)	Reliable demo, low compute
Phase 2 (Sem 8)	Eye-gaze estimation + dwell calibration	Higher precision, stronger paper
Both	Dwell 0.5-1.0 s before EMG counts	Blocks accidental glances

5. Demo examples

Clear Act

- User looks at lamp for 0.8 s -> target = lamp
- EMG spike -> ACT lamp ON via relay hub

Abstain: look only

- User glances at fan, no EMG -> ABSTAIN (looking != wanting)

Abstain: EMG without gaze

- Muscle tense while talking, no device gaze -> ABSTAIN

Abstain: ambiguous

- Lamp and fan both near gaze / low confidence -> ABSTAIN

6. Algorithms locked

Stage	Phase 1	Phase 2
Objects	YOLO11s / YOLOv8s (or RT-DETR)	Same + optional open-vocab assist
Gaze	MediaPipe Face Mesh / head pose	Eye-gaze model (iris / L2CS-style)
EMG	RMS/MAV threshold or small TCN	Calibrated volition score
Policy	gaze AND EMG -> Act else Abstain	Learned fusion + FAR-tuned abstention
IoT	MQTT + ESP32 + relay demo hub	Same + caregiver XAI dashboard
Metric	FAR + intention accuracy (pilot)	FAR + F1 + ablations + paper

7. Delta over base paper (EMBC 2024)

Point	Chamberland et al.	NeuroShift
Vision role	Validate prosthesis grips	Detect appliances + gaze-select target
EMG role	Classify gesture	Confirm toggle intent only
Selector	Hand near object (MediaPipe Hands)	Head/eye gaze (primary)
Output	Grip class	Device Act or Abstain
Safety	Fewer false gestures	Two-factor + FAR metric

8. Viva one-liner

NeuroShift uses the camera to detect appliances and which one the user is looking at; the EMG sensor then confirms whether they truly want to toggle it. If either signal is unclear, the system abstains to reduce false actuations.

Full markdown: research/07_gaze_emg_pipeline.md